

Average Case Time Complexity Analysis of the Improved Sequential Search Algorithm

Zhesheng Xu
Student ID: 42353012

March 5th 2025

Analysis of Average Time Complexity

Assume that the list L has length n , and the target element x is equally likely to appear at any position in L or not be in L at all.

Successful Search Case (x is in L)

If x is in L , it can appear at any position i with equal probability $\frac{1}{n}$. Finding x at position i requires i comparisons. Therefore, the expected number of comparisons for a successful search is:

$$A(n) = \sum_{i=1}^n i \cdot \frac{1}{n}$$

Using the formula for the sum of the first n natural numbers:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

we substitute this into the equation:

$$A(n) = \frac{1}{n} \cdot \frac{n(n+1)}{2}$$

Canceling n :

$$A(n) = \frac{n+1}{2}$$

Unsuccessful Search Case (x is not in L)

If x is not in L , the algorithm stops at the first element greater than x . Assuming x is uniformly distributed across $n+1$ possible intervals (including before and after the list), we compute the expected number of comparisons as follows:

The comparisons required for each of the first n elements are $1, 2, \dots, n$. If x is greater than all elements in L , the algorithm makes n comparisons. Since there are $n + 1$ possible positions for x , the expected number of comparisons is:

$$A(n) = \frac{1 + 2 + \dots + n + n}{n + 1}$$

Using the formula for the sum of the first n natural numbers:

$$1 + 2 + \dots + n = \frac{n(n + 1)}{2}$$

we substitute this into the equation:

$$A(n) = \frac{\frac{n(n+1)}{2} + n}{n + 1}$$

Factor out n in the numerator:

$$A(n) = \frac{n(n + 1) + 2n}{2(n + 1)} = \frac{n(n + 3)}{2(n + 1)}$$

Overall Average Time Complexity

Assume that x is in L with probability p and is not in L with probability $1 - p$. The expected number of comparisons is:

$$T(n) = p \cdot \frac{n + 1}{2} + (1 - p) \cdot \frac{n(n + 3)}{2(n + 1)}$$

If we assume $p = 0.5$, then:

$$T(n) = 0.5 \cdot \frac{n + 1}{2} + 0.5 \cdot \frac{n(n + 3)}{2(n + 1)}$$

Simplifying:

$$T(n) = \frac{2n^2 + 5n + 1}{4(n + 1)}$$

For large n , this approximates to:

$$T(n) \approx \frac{n}{2}$$

Thus, the **average case time complexity of the improved sequential search algorithm** is $O(n)$.